Code: DMB O2 *Paper ID: 13442*

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER –III) END TERM EXAMINATION, APRIL -2024

BASIC SQL FOR DATA ANALYTICS

[Time: 3 Hours] ***[Maximum Marks: 60]***

Following are the course outcomes of the subject: - Basic SQL for Data Analytics

CO	Course Outcome (CO)
CO1	Understand the fundamental elements of relational database management systems. Will be clear with the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL.
CO2	Design ER-models to represent simple database application scenarios. Enhance the knowledge of the processes of Database Development and Administration using SQL.
CO3	Convert the ER-model to relational tables, populate relational databases and formulate SQL queries on data. Execute the commanding functions on various DB platforms.

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 200 words)

Q.1	Marks	CO
(A)	**4**	CO – 1
(B)	**4**	CO – 1
(C)	**4**	CO – 1
(D)	**4**	CO – 1
(E)	**4**	CO – 1

SECTION-B

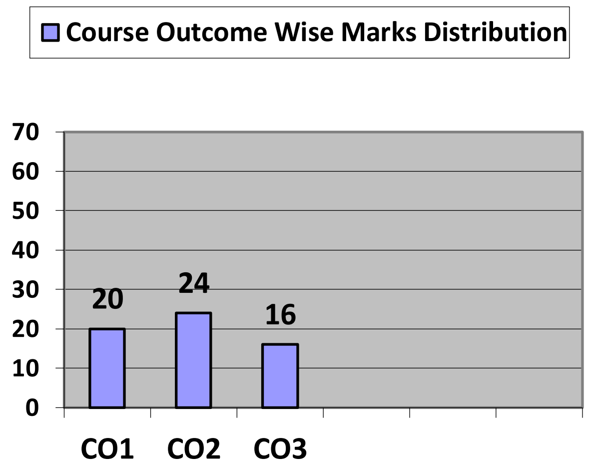
Attempt all questions out of the following: (Long Answer Type Questions)

Q. No	Marks	CO
2 (A)	**6**	CO – 2
2 (A)	**6**	CO – 2
2 (B)	**6**	CO – 2
2 (B)	**6**	CO – 2
3 (A)	**6**	CO – 2
3 (A)	**6**	CO – 2
3 (B)	**6**	CO – 2
3 (B)	**6**	CO – 2

SECTION-C

Attempt all parts of the following: (Caselet / Numerical)

Q.4	Marks	CO
4 (A)	**8**	CO – 3
4 (B)	**8**	CO – 3



Moderated By **Head- IQAC** **Dean - Examinations**

Printed Pages: 2

Roll No: ☐☐☐☐☐☐☐☐☐☐ ***Subject Code: DMB 02***

Paper ID: 13442

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)

(TRIMESTER – III) END TERM EXAMINATION, APRIL - 2024

BASIC SQL FOR DATA ANALYTICS

[Time: 3 Hours] ***[Maximum Marks: 60]***

INSTRUCTIONS:

(i) There are THREE sections in this question paper.
(ii) Attempt questions from each section as per the directions.
(iii) Make suitable assumptions wherever necessary.

SECTION-A
(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following: ***(5×4=20)***

(a) Explain Data Abstraction in DBMS along with its levels.
(b) Describe Instances and Schemas in DBMS, support with example.
(c) Different types of keys with exampls.
(d) Create Client_master with the following fields (ClientNO, Name, Address, City, State, bal_due)
1. Insert five records.
2. Find the names of clients whose bal_due> 5000.
3. Change the bal_due of ClientNO " C123" to Rs. 5100
4. Change the name of Client_master to Client12 .

(e) Create Sales table with the following fields (Sales No, Salesname, Branch, Salesamount, DOB)
1. Insert five records
2. Calculate total salesamount in each branch.
3. Display all the salesmen, DOB who are born in the month of December.
4. Display the name and DOB of salesman in alphabetical order of the month.

SECTION-B
(Long Answer Type Questions)

Q.2. Attempt all questions of the following ***(4×6=24)***

(a) What is a Database Management System and how is it different from a File System?

OR

(a) An Enterprise wishes to maintain a database to automate its operations. Enterprise is divided into certain departments and each department consists of employees. The following two tables describes the automation schemas
Dept (deptno, dname, loc)
Emp (empno, ename, job, mngr, hiredate, sal, deptno)
Create Table and insert five records.
1) Update the employee salary by 15%, whose experience is greater than 10 years.
2) Delete the employees, who completed 30 years of service.
3) Display the manager who is having maximum number of employees working under him.

(b) Explain the Data Model in DBMS and what are its types?

OR

(b) Describe attributes in DBMS along with its types (with example).

Q.3. *(a)* Define an ER Model in detail. Illustrate with examples.

OR

(a) Explain with examples different types of relationships in DBMS

(b) Difference between SQL and NoSQL.

OR

(b) MAC prides itself on having up-to-date information on the processing and current location of each shipped item.

a. To do this, MAC relies on a company-wide information system. Shipped items are the heart of the MAC product tracking information system.
b. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date.
c. Shipped items are received into the UPS system at a single retail center.
d. Retail centers are characterized by their type, uniqueID, and address.
e. Shipped items make their way to their destination via one or more standard MAC transportation events (i.e., flights, truck deliveries).
f. These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute.

Please **create an Entity Relationship diagram** that captures this information about the MAC system. Be certain to indicate identifiers and cardinality constraints.

SECTION-C

Q.4. Attempt all parts of the following: ***(2×8=16)***

4(a) Create a Supplier table as shown below : (for questions from 1 to 6)

Sup_No (Primary Key)	Sup_Name	Item_Supplied	Item_Price	City
S1	Suresh	Keyboard	400	Hyderabad
S2	Kiran	Processor	8000	Delhi
S3	Mohan	Mouse	350	Delhi
S4	Ramesh	Processor	9000	Bangalore
S5	Manish	Printer	6000	Mumbai
S6	Srikanth	Processor	8500	Chennai

1. Write sql query to display Supplier numbers and Supplier names whose name starts with 'R'
2. Write sql query to display the name of suppliers who supply Processors and whose city is Delhi.
3. Write sql query to increase the price of Keyboard by 200.
4. Write sql query to display supplier numbers, Suplier names and itemprice for suppliers in delhi in the ascending order of itemprice.
5. Write sql query to add a new column called CONTACTNO.
6. Write sql query to display the records of suppliers who supply items other than Processor or Keyboard.

4(b) Create the table called EmpDetails with the below mentioned details. (For questions from 1 to 6)

1. Write sql query to display all the employees whose designation is Programmer.
2. Write sql query to display employees who have joined after 2014.
3. Write sql query to display the TOTAL salary of all the employees whose designation is programmer.
4. Write sql query to display the details of the employee with highest experience.
5. Write sql query to display the details of the employees whose name contains'ee'.
6. Write sql query to increase the salaries of employees by 5000 whose designation is DBA.

Eid (Primary Key)	Ename	DOB	Designation	Salary	DOJ
E101	Suma	29-Dec-89	Designer	20000	01-Apr-10
E102	Amit	10-Jan-95	Programmer	25000	18-Feb-18
E103	Payal	15-Aug-85	Tester	35000	13-Jun-11
E104	Kiran	20-Apr-90	Programmer	40000	7-Mar-14
E105	Meenal	29-May-83	DBA	50000	9-Dec-11
E106	Sheila	1-May-70	Analyst	60000	25-Sep-18
E107	Swamy	13-Jan-85	Programmer	45000	14-Feb-16
E108	Sushma	22-Dec-76	DBA	45000	31-Jan-12